

Algebraic Descent in Mixed-Dimensional Gaussian Traces: A Period Ring Formulation

Abstract

We construct an analytic invariant, the *Mixed Gaussian Trace* $\mathcal{Z}(p, q)$, defined as the tensor product of the continuous Gaussian measure on $\mathbb{R}^p$ and the discrete heat trace on the flat torus $\mathbb{T}^q$, evaluated at the modular fixed point. By framing the trace within the Kontsevich-Zagier ring of periods and invoking Chudnovsky's theorem on the algebraic independence of π and $\Gamma(1/4)$, we show that $\mathcal{Z}(p, q)$ undergoes a formal algebraic descent if and only if $2p = 3q$. At the fundamental integer solution ($p = 3, q = 2$), the canonical period π precisely cancels, and the invariant drops into the purely elliptic period ring governed by the complex multiplication of the lemniscate curve.

1. The Mixed Spectral Trace

We define a strict analytic invariant via spectral geometry. Let $\Delta_{\mathbb{T}^q}$ be the positive Laplace-Beltrami operator on the standard flat torus $\mathbb{T}^q = \mathbb{R}^q / \mathbb{Z}^q$. The eigenvalues of the Laplacian are $\lambda_n = 4\pi^2 \|n\|^2$ for $n \in \mathbb{Z}^q$. The trace of the heat kernel at spectral time $t > 0$ is the multidimensional Jacobi theta function:

$$K_{\mathbb{T}^q}(t) = \mathrm{Tr} \left(e^{-t\Delta_{\mathbb{T}^q}} \right) = \sum_{n \in \mathbb{Z}^q} e^{-4\pi^2 \|n\|^2 t} \quad (1)$$

We evaluate this trace at the specific spectral time $t = \frac{1}{4\pi}$, corresponding to the self-dual point $\tau = i$ under the modular transformation $\tau \mapsto -1/\tau$. This yields exactly $K_{\mathbb{T}^q} \left(\frac{1}{4\pi} \right) = \theta_3(e^{-\pi})^q$. Concurrently, we evaluate the standard L^2 -normalized Gaussian measure over $\mathbb{R}^p$.

Definition 1 (The Mixed Gaussian Trace). *Let $\mathcal{Z}(p, q)$ be the tensor product of the continuous Gaussian measure on $\mathbb{R}^p$ and the discrete heat trace on $\mathbb{T}^q$ at $t = \frac{1}{4\pi}$:*

$$\mathcal{Z}(p, q) = \left(\int_{\mathbb{R}^p} e^{-\|x\|^2/2} d^p x \right) \times \mathrm{Tr} \left(e^{-\frac{1}{4\pi} \Delta_{\mathbb{T}^q}} \right) \quad (2)$$

2. The Chowla-Selberg Formula and Periods

We contextualize the evaluation of this trace within the theory of periods. The self-dual torus evaluated at $\tau = i$ corresponds to the elliptic curve $E : y^2 = x^3 - x$, which has Complex Multiplication (CM) by the Gaussian integers $\mathbb{Z}[i]$.

By the Chowla-Selberg formula, the periods of CM elliptic curves are evaluated analytically in terms of the Euler Gamma function at rational arguments. Specifically, the trace at $\tau = i$ resolves exactly to:

$$\theta_3(e^{-\pi}) = \frac{\Gamma(1/4)}{2^{1/2}\pi^{3/4}} \quad (3)$$

Taking the product across arbitrary dimensions (p, q) yields the exact closed form:

$$\mathcal{Z}(p, q) = (2\pi)^{\frac{p}{2}} \cdot \left(\frac{\Gamma(1/4)}{2^{1/2}\pi^{3/4}} \right)^q = 2^{\frac{p-q}{2}} \cdot \pi^{\frac{2p-3q}{4}} \cdot \Gamma\left(\frac{1}{4}\right)^q \quad (4)$$

This expression highlights that $\mathcal{Z}(p, q)$ is fundamentally a mixed state of two distinct transcendental periods: the circular period π and the lemniscatic period associated with $\Gamma(1/4)$.

3. Algebraic Descent via Chudnovsky's Theorem

In 1984, G. V. Chudnovsky proved a profound theorem in transcendental number theory: π and $\Gamma(1/4)$ are algebraically independent over $\mathbb{Q}$. Consequently, no non-trivial polynomial relation exists between them. This dictates that the fractional exponent of π in Equation 4 cannot be algebraically simplified, absorbed, or rationalized by any power of $\Gamma(1/4)$.

We define the *transcendental π -weight* as the rational exponent $\frac{2p-3q}{4}$. Setting this weight identically to zero is strictly mathematically necessary for the trace to algebraically descend from a mixed transcendental state over $\mathbb{Q}(\pi, \Gamma(1/4))$ into a pure algebraic extension of the lemniscate period.

Theorem 1 (Period Descent Condition). *The Mixed Gaussian Trace $\mathcal{Z}(p, q)$ is algebraically independent of the circular period π if and only if the dimensional parameters satisfy the linear Diophantine equation:*

$$2p = 3q \quad (5)$$

4. The (3,2) Elliptic Trace

The minimal non-trivial positive integer combination satisfying this period descent is $p = 3$ and $q = 2$. When evaluated at this dimensional ratio, the transcendental π -weight vanishes

entirely ($\pi^0 = 1$):

$$\begin{aligned} \mathcal{Z}(3, 2) &= 2^{\frac{3-2}{2}} \cdot \pi^0 \cdot \Gamma\left(\frac{1}{4}\right)^2 \\ \implies &\boxed{\mathcal{Z}(3, 2) = \sqrt{2} \cdot \Gamma\left(\frac{1}{4}\right)^2 \approx 18.5899 \dots} \end{aligned} \tag{6}$$

Remark 1. *The initial heuristic identity $G^* = \frac{\Gamma(1/4)^2}{\pi\sqrt{2}}$ evaluated a 1-dimensional continuous measure against a 2-dimensional toroidal trace ($p = 1, q = 2$). Under this rigorous formal framework, the appearance of π in the denominator of G^* is revealed to be an uncanceled period weight mandated by Chudnovsky's theorem: because $2(1) \neq 3(2)$, the invariant fails the algebraic descent condition, and the period π is mathematically forced to remain. By mapping $\mathcal{Z}(3, 2)$, we isolate the exact arithmetic resonance where the Euclidean volume measure perfectly commensurates with the modular weight of the heat trace.*